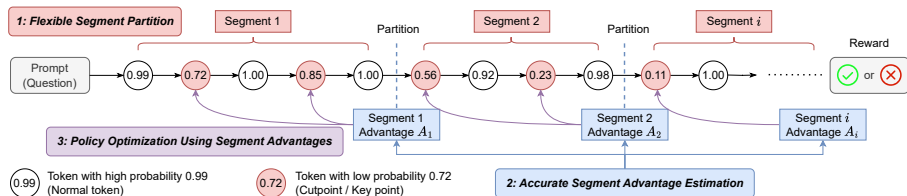


Segment Policy Optimization: Segment-Level Credit Assignment for RL in LLMs

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SPO Overview: Segment-Level Credit Assignment



Key idea: move from trajectory-level or token-level to segment-level.

- Partition $\mathbf{y} = [y_1, \dots, y_T]$ into contiguous segments $\{\text{seg}_k\}_{k=1}^K$.
- Define a segment-advantage as an incremental change in V across segment boundaries.
- Estimate V via Monte Carlo (critic-free), then optimize a clipped surrogate with masks.

Segment and Segment-Advantage

Segment k with boundaries t_k and t_{k+1} :

$$\text{seg}_k = [y_{t_k}, \dots, y_{t_{k+1}-1}], \quad \mathbf{y} = [\text{seg}_1, \dots, \text{seg}_K].$$

Segment-level advantage (definition):

$$A_k^{\text{seg}} := V(s_{t_{k+1}}) - V(s_{t_k}) = V([\mathbf{x}, \text{seg}_1, \dots, \text{seg}_k]) - V([\mathbf{x}, \text{seg}_1, \dots, \text{seg}_{k-1}]).$$

Remarks:

Critic-free MC estimation is viable since rewards are Bernoulli; $\text{Var}(\hat{V}) \leq 0.25/N$.

Partition Strategies (Mathematical View)

Two strategies supported by the framework:

- Fixed token count: split every M tokens.
- Cutpoint-based (short CoT): define cutpoints $\mathcal{U}_\theta = \{t < T \mid \pi_\theta(y_t \mid s_t) < \rho\}$; given K segments, solve

$$\min_{\{t_k\}} \sum_{k=1}^K |\mathcal{U}_\theta \cap [t_k, t_{k+1})|^2$$

whose minimizer evenly distributes cutpoints across segments, i.e., $|\mathcal{U}_\theta \cap [t_k, t_{k+1})| = |\mathcal{U}_\theta|/K$.

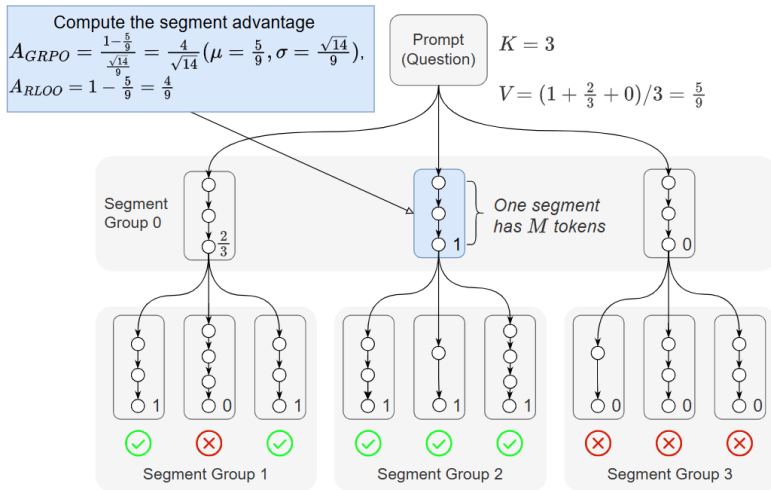
For boundary state s_{t_k} , sample N rollouts $\tau_{t_k}^{(j)} \sim \pi_{\theta}(\cdot \mid s_{t_k})$ and estimate

$$\hat{V}(s_{t_k}) = \frac{1}{N} \sum_{j=1}^N \mathcal{R}(\mathbf{x}, [\mathbf{y}_{< t_k}, \tau_{t_k}^{(j)}]), \quad \hat{A}_k^{\text{seg}} = \hat{V}(s_{t_{k+1}}) - \hat{V}(s_{t_k}).$$

Training objective with probability masks (focus on low-probability tokens):

$$\mathcal{J}_{\text{SPO}}^{\text{chain}}(\theta) = \mathbb{E} \left[\frac{1}{Z} \sum_{k=1}^K \sum_{t=t_k}^{t_{k+1}-1} M_t \cdot \min(r_t \hat{A}_k^{\text{seg}}, \text{clip}(r_t, 1 - \epsilon, 1 + \epsilon) \hat{A}_k^{\text{seg}}) \right. \\ \left. - \beta D_{\text{KL}}(\pi_{\theta} \parallel \pi_{\text{ref}}) \right], \quad M_t = \mathbb{I}(\pi_{\theta_{\text{old}}}(a_t \mid s_t) < \rho).$$

Tree-Based MC: Reusing Samples Efficiently



Tree-Based MC: Reusing Samples Efficiently

Model sampling as a tree. For node n with history $\text{hist}(n)$ and segment $\text{seg}(n)$ of M tokens:

$$\hat{V}(n) = \begin{cases} \mathcal{R}(\mathbf{x}, \text{hist}(n)), & n \text{ is leaf} \\ \frac{1}{|\text{Ch}(n)|} \sum_{n' \in \text{Ch}(n)} \hat{V}(n'), & \text{otherwise} \end{cases}$$

Advantage within sibling group:

$$\hat{A}(n) = \hat{V}(n) - \text{mean}_{n' \in \text{Sib}(n)} \hat{V}(n') \quad \text{or} \quad \frac{\hat{V}(n) - \text{mean}}{\text{std}}.$$

Training objective (masked tokens, non-zero advantages):

$$\mathcal{J}_{\text{SPO}}^{\text{tree}}(\theta) = \mathbb{E} \left[\frac{1}{Z} \sum_{n: \hat{A}(n) \neq 0} \sum_{t=1}^{|\text{seg}(n)|} M_{n,t} \cdot \min(r_{n,t} \hat{A}(n), \text{clip}(r_{n,t}, 1 - \epsilon, 1 + \epsilon) \hat{A}(n)) \right. \\ \left. - \beta D_{\text{KL}}(\pi_{\theta} \parallel \pi_{\text{ref}}) \right].$$

Why Segment-Level Works (Math Intuition)

- Signal-to-Noise: $\hat{V}$ is Bernoulli-MC; segment-level differences ΔV provide concentrated signals at boundary changes.
- Bias–Variance: token-level needs a critic (bias risk); trajectory-level has sparse credit; segment-level balances both.
- Masking focuses gradient on low-probability tokens that drive ΔV , reducing loss on near-deterministic tokens.
- Tree reuses samples: bottom-up $\hat{V}$ aggregation reduces sample complexity while supplying many training segments.